Designing a DC-DC Buck Converter and a Closed-loop Voltage control

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1. Brief Description of Theory/Methodology

This project focuses on the modeling, control, and simulation of a buck converter operating under various load and input voltage conditions. The buck converter steps down a higher DC input voltage to a lower DC output voltage. The system can operate in Continuous Conduction Mode (CCM), where the inductor current never reaches zero, or Discontinuous Conduction Mode (DCM), where it does. A PI control strategy is implemented to regulate the output voltage through cascaded inner current and outer voltage control loops.

2. Detailed Description of the Matlab/Simulink Model

The Simulink model includes the essential components of the buck converter such as IGBT switch (IRG4PC50W), diode (MUR3060WT), inductor (1.2 mH), and capacitor (1.6 µF). The control structure consists of a current controller and a voltage controller implemented using PI blocks. A PWM signal generated from the current loop controls the IGBT. The input voltage source is configured to perform step changes (e.g., from 140 V to 180 V), and the load is modeled as a variable resistor to simulate no-load, full-load, and light-load (DCM) conditions. Simscape blocks and controlled voltage sources are used to model physical behavior realistically.

The main components used in the converter are listed below, along with their models and specifications selected based on datasheet availability.

Component	Model	Key Specs
IGBT	IRG4PC50W	600V / 45A, TO-247 package, Fast switching Available at: RS Components, TME, Farnell
Diode	MUR3060WT	600V / 30A, Ultra-fast recovery Available at: RS Components, TME
Inductor	Würth 744355120	1.2mH, 14.5A saturation current Available at: RS Components, Farnell, Mouser
Capacitor	WIMA MKP10 1.6µF 250V	1.6uF, 250V, Polypropylene Film, Low ESR Available at: RS Components, Farnell, TME.eu

3. Calculation Results

The output voltage is set to 110 V. The duty cycle at nominal input (160 V) is calculated as D = 0.6875. The boundary between CCM and DCM is defined by half the inductor current ripple:
$\Delta I_L = (Vin - Vout)/L * D * T$, with fs = 40 kHz (T = 25 µs) ⇒ $\Delta I_L \approx 0.715$ A. $I_{border} = \Delta I_L / 2 \approx 0.358$ A.
Below this current level, the converter operates in DCM.
Inductor: Inductance is calculated for three input voltage levels (Vin = 140 V, 160 V, 180 V) using Vout = 110 V and switching frequency fs = 40 kHz.
Target inductor current ripple: dIL = 0.05 x 22.7 A ~= 1.136 A (5% of full-load current)
Vin = 140 V: D = 110/140 = 0.7857, T = 25 us -> Lmin ~= (140-110) x 0.7857 x 25e-6 / 1.136 = 0.519 mH
Vin = 160 V: D = 110/160 = 0.6875, T = 25 us -> Lmin ~= (160-110) x 0.6875 x 25e-6 / 1.136 = 0.756 mH
For Vin = 180 V: D = Vout/Vin = 110/180 = 0.6111, fs = 40 kHz (T = 25 us)
Required inductance (target dIL = 1.136 A):
Lmin ~= (Vin - Vout) x D x T / dIL = (180 - 110) x 0.6111 x 25e-6 / 1.136 = 0.942 mH
Selected: L = 1.2 mH (commercial value). With fs = 40 kHz, the ripple requirement is satisfied for Vin = 140-180 V.
Capacitor design for output ripple: dVout <= 10% x Vout = 11 V (worst case Vin = 180 V, D = 0.6111): Cmin = dIL x (1 - D) x T / dVout
Cmin ~= 1.136 x 0.3889 x 25e-6 / 11 = 1.00 uF -> Selected: C = 1.6 uF (WIMA MKP10).

4. Simulation Results

Figure 1 shows the final Simulink implementation used for simulation, including the control loops, pulse generation, and measurement blocks.

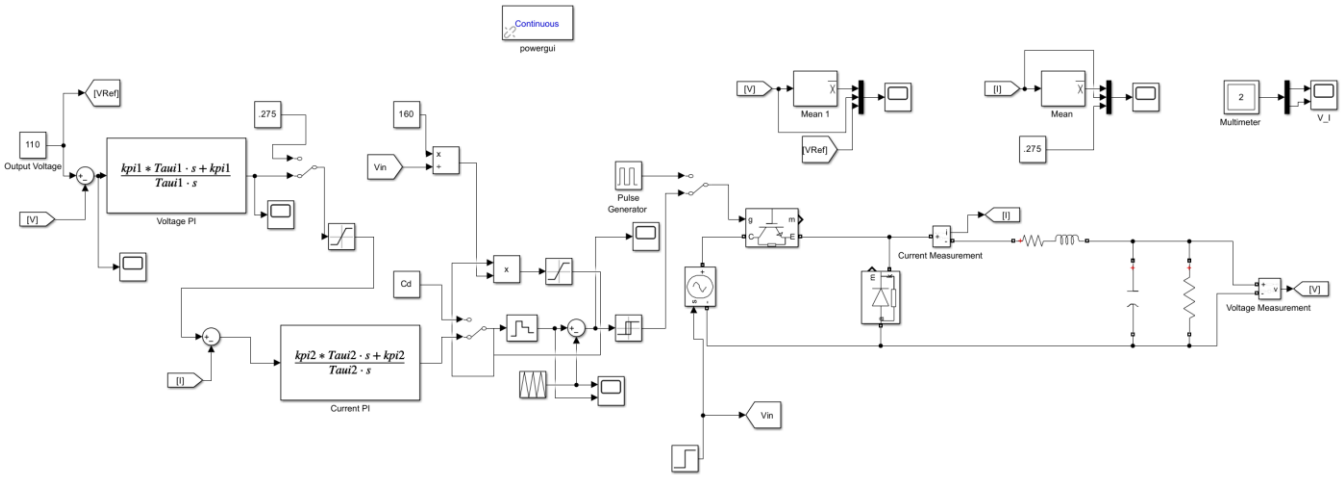


Figure 1: Simulink power stage + control system for Vin 160.

Control System Behavior

Figures 2 to 6 show Bode plots and time responses. Since Vin changes the required duty, a simple feedforward term (Dff = Vref/Vin) is added and scaled the PI output by 1/Vin so the response stays similar when Vin varies, verifying the stability and accuracy of the current and voltage regulation.

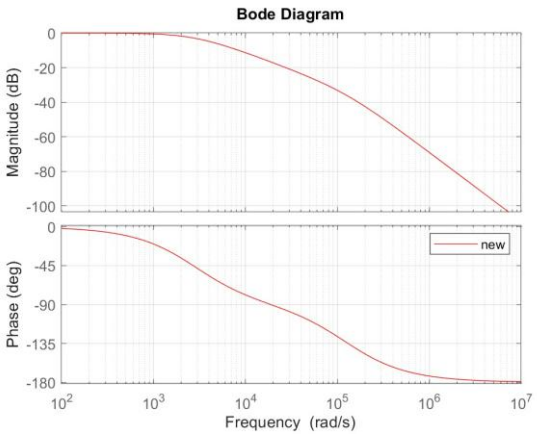


Figure 2: Open-loop Bode diagram of the system.

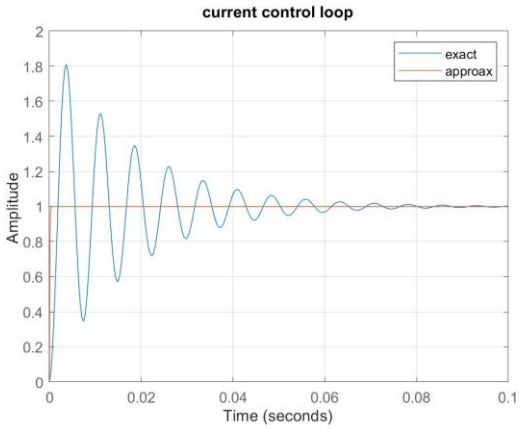


Figure 4: Time-domain response of the current control loop.

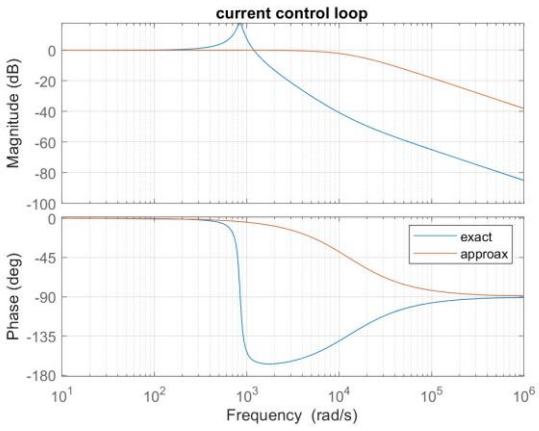


Figure 3: Bode diagram of the current control loop (exact vs approximate).

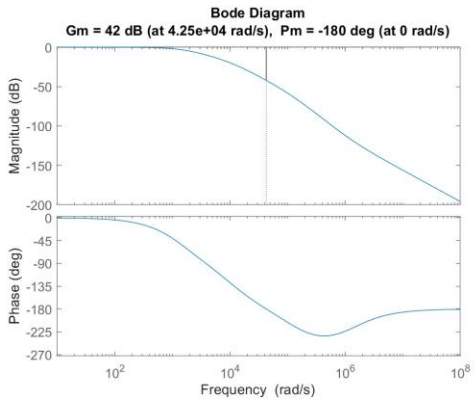


Figure 5: Bode diagram showing gain and phase margins.

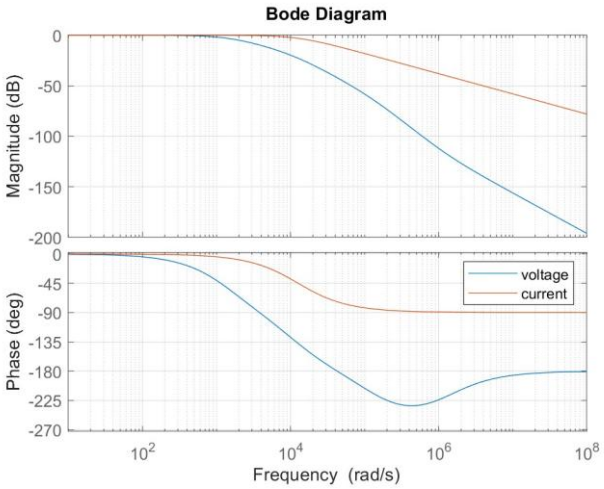


Figure 6: Comparison of voltage and current loop Bode diagrams.

Steady-State Response with Different Input Voltages

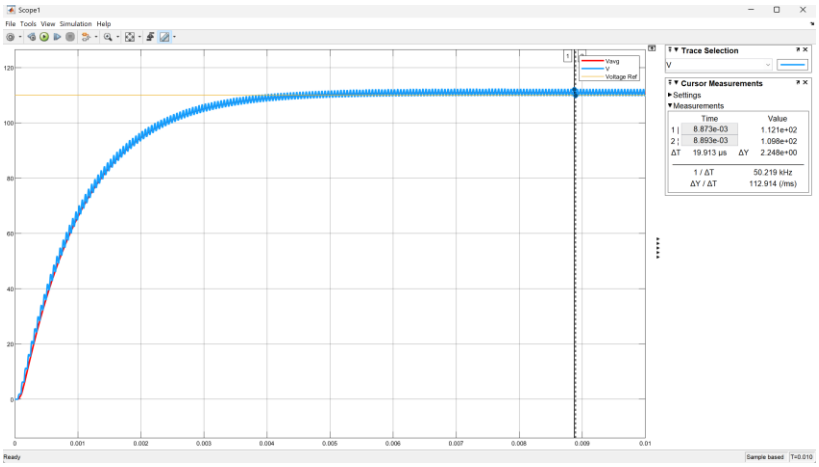


Figure 7. Output Voltage Response for Vin = 140 V (Full Load, RL = 4.84 Ω).

This figure shows the transient response of the output voltage when the input is 140 V. The converter successfully regulates the output to the reference value of 110 V with minimal overshoot and steady convergence, indicating proper operation in continuous conduction mode.

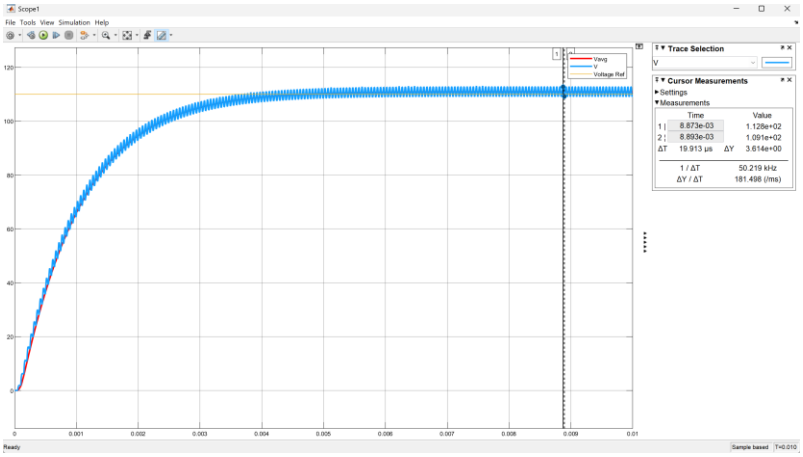


Figure 8: Steady-state output voltage waveform of the buck converter at Vin = 160 V (Full Load, RL = 4.84 Ω).

With a 160 V input, the buck converter maintains a regulated output close to the 110 V reference. While the ripple is slightly higher than at 140 V, the overall response remains stable and within acceptable limits.

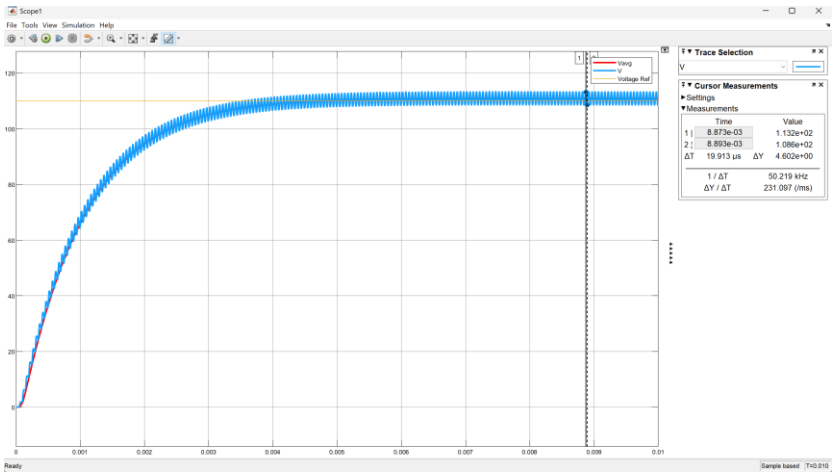


Figure 9: Steady-state output voltage waveform of the buck converter at Vin = 180 V (Full Load, RL = 4.84 Ω).

At 180 V input, the converter stabilizes the output voltage at 110 V. The output voltage ripple is approximately **4.18%**, showing slightly increased ripple due to higher input voltage, but the response remains stable and within acceptable bounds.

For the input voltages of 140 V, 160 V, and 180 V, the output voltage successfully settles around the reference value of 110 V. As the input voltage increases, the response becomes slightly faster, and the output ripple increases marginally, reaching approximately 4.18% at 180 V, but remains within acceptable limits.

Step Response at No Load (R = 10 kΩ)

With minimal output current (~11 mA), the system demonstrated a slow voltage rise, reaching only about 12 V. This is due to limited energy transfer and lack of load-induced current flow. R = 10 kΩ was chosen as a practical approximation of a no-load condition.

Step Response at Full Load (R = 4.84 Ω)

The converter responded quickly and accurately, with the output voltage reaching and stabilizing at 110 V. There was minimal overshoot and nearly zero steady-state error, confirming effective control.

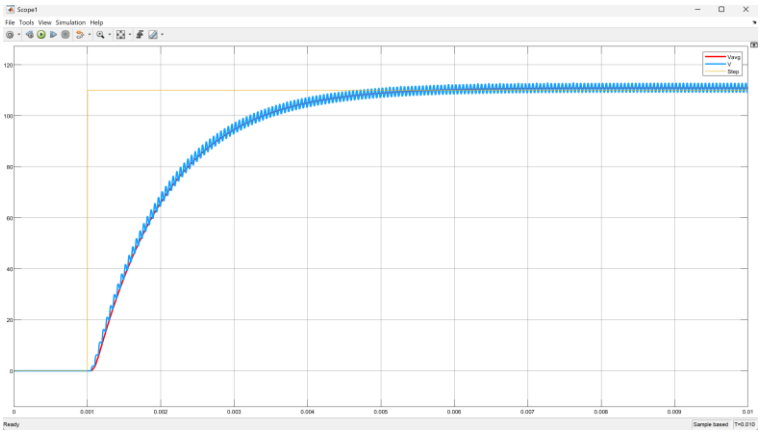


Figure 10: Output Voltage Step Response at Vin = 160 V and Full Load (R = 4.84 Ω).

Step response of the converter at full load (R = 4.84 Ω). After the step input from 0 V to 110 V at 1 ms, the output voltage rises smoothly and stabilizes near the reference with minimal overshoot and acceptable ripple.

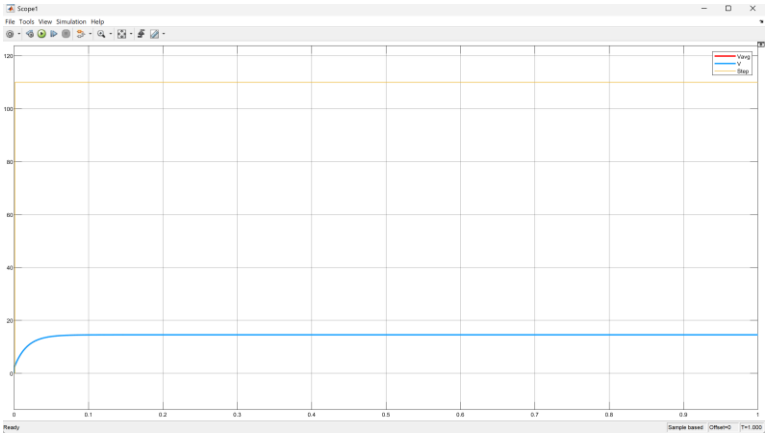


Figure 11: Output Voltage Step Response at No Load (R = 10 kΩ).

The converter's output fails to reach the reference voltage of 110 V even after 1 second due to the extremely low load current. This illustrates that under no-load conditions, the converter is unable to deliver sufficient control effort to build up and regulate the output voltage effectively.

In conclusion we can see that under full load conditions, the transient response is fast and the output voltage accurately reaches the reference value. In contrast, under no-load conditions, due to the very low current, the output fails to reach the reference even after one second, indicating ineffective control.

Step-Up in Input Voltage (140 V → 180 V) under DCM

To calculate the boundary resistance (R) between Continuous Conduction Mode (CCM) and Discontinuous Conduction Mode (DCM), the following analytical formula is used:

R_boundary = (2 × L × f_s) / (1 - D)

Where: L = 1.2 mH (inductance), f_s = 40000 Hz (switching frequency), D = 110 / 160 = 0.6875 (duty cycle)

By substituting the values into the formula: ***R_boundary = (2 × 1.2 × 10^-3 × 4 × 10^4) / (1 - 0.6875) = 96 / 0.3125 = 307.2 Ω but in terms of current we have:***

Condition	*Load*	*R (Ω)*
CCM	*1.1 A*	*100 Ω*
CCM	*0.44 A*	*250 Ω*
CCM/DCM	*0.358 A*	*307 Ω*
DCM	*0.275 A*	*400 Ω*
DCM (deep)	*0.11 A*	*1000 Ω*

This calculation gives a boundary resistance of approximately 307 Ω. To ensure Discontinuous Conduction Mode (DCM) operation, a resistance upper than this value is chosen. According to above table For the simulation, R = 420 Ω and A = 0.275A is selected to guarantee DCM operation.

With a 420 Ω load, the converter operated in DCM. When Vin steps from 140 V to 180 V at t = 1 s, the output exhibited a noticeable overshoot. To reduce this effect, the duty command is limited (0 to 1), the current reference is limited, and a simple Vin feedforward term (Dff = Vref/Vin) is used so the controller does not request an excessive duty during the transient.

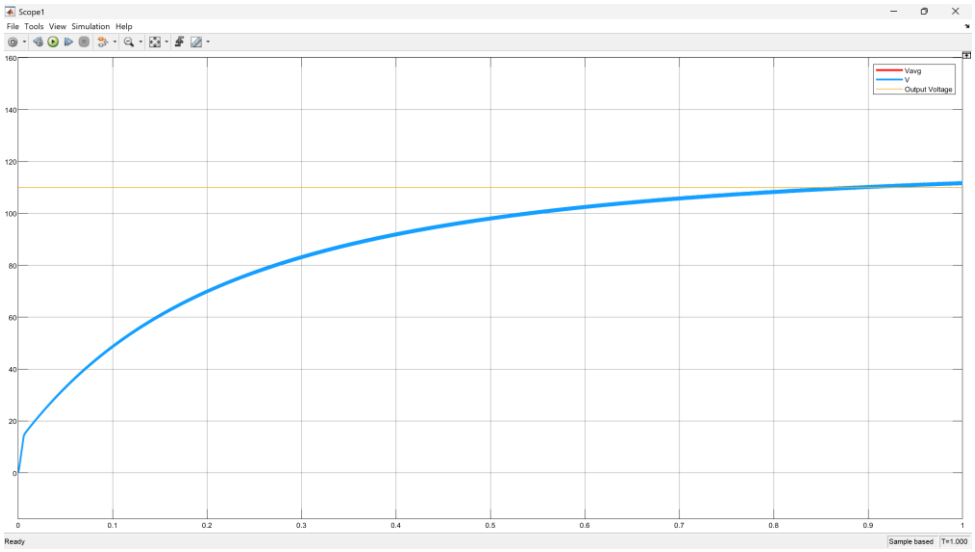


Figure 12: Output Voltage Response to a Step-Up in Vin (140 V → 180 V) under DCM (R = 420 Ω).

The output voltage waveform shows characteristics of DCM operation, including light overshoot and oscillations. These occur because the inductor current drops to zero during part of the switching cycle, which typically happens at light load (high resistance or low current demand).

Step-Down in Input Voltage (180 V → 140 V) under DCM

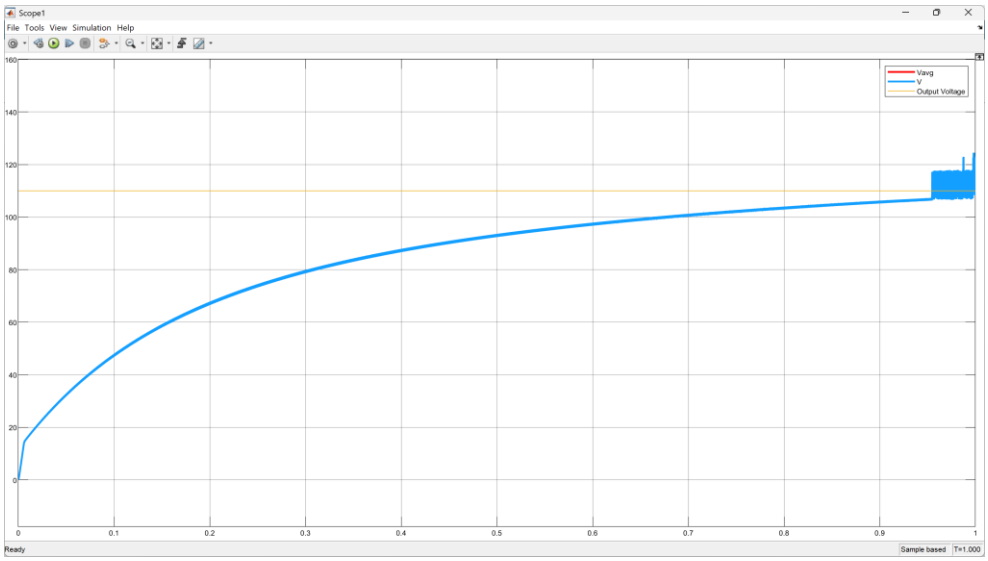


Figure 13: Output Voltage during 180 V to 140 V Step-Down under DCM (R = 420 Ω).

The output voltage rises toward the reference value during start-up with limited overshoot. After the input voltage steps down from 180 V to 140 V, a transient drop in the output voltage is observed, followed by a gradual return toward the reference value. This behavior is expected under light-load conditions (high resistance), where the converter may enter discontinuous conduction mode (DCM)

Final Conclusion

In this project, a buck converter with variable input voltage was designed, analyzed, and simulated under various load and line conditions. The converter demonstrated stable performance in both Continuous and Discontinuous Conduction Modes, with acceptable ripple and transient response. Control strategies, including current and voltage feedback loops, were implemented and evaluated through Bode plots and time-domain simulations. Overall, the system successfully met the regulation requirements and responded effectively to dynamic changes, validating the design approach.